YASH VARIK

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building production-grade LLM systems, including Corrective RAG pipelines, LangGraph agentic workflows, and custom MCP servers. Skilled in evaluating and monitoring LLM applications using RAGAS and LangSmith, with guardrails for safe, reliable outputs. Currently pursuing M.Sc. Computer Science (2025–2027) with a 9.0 CGPA undergraduate record. Focused on deploying end-to-end agentic AI applications on AWS using Python, FastAPI, and Docker.

EXPERIENCE

AI Engineer Intern — Sell Instant *April 2026 – Present*

Mumbai, India

- Build and deploy LLM-powered features for the core product using LangChain and FastAPI, shipping changes directly to production.
- Design agentic AI workflows with LangGraph to automate multi-step task handling and reasoning, reducing manual intervention in repetitive workflows.
- Collaborate with the engineering team to integrate AI modules into the core platform through REST APIs.

PROJECTS

C-RAG — Corrective Retrieval Augmented Generation

Advanced RAG architecture with multi-stage document filtering and sentence-level context selection

- Built an LLM-based chunk-filtering layer that strips retrieved documents down to only the most query-relevant sentences before generation, reducing irrelevant context passed to the LLM.
- Added a corrective re-ranking and validation layer on top of standard RAG retrieval, evaluated with RAGAS (faithfulness, answer relevancy, context precision) against a baseline RAG pipeline.

Stack: Python, LangChain, FAISS, OpenAI API, RAGAS

LangGraph Agentic Chatbot with Long-Term Memory

Stateful AI agent with persistent memory across sessions using LangGraph

- Built a multi-step reasoning agent with tool-calling, state management, and memory that persists across conversations, enabling context-aware follow-up interactions.
- Designed a subgraph architecture in LangGraph to modularize complex agent workflows, making the system easier to extend and debug.
- Used LangSmith to trace and debug multi-step agent runs, and Guardrails AI to validate and constrain LLM outputs for reliability.

Stack: Python, LangGraph, LangChain, SQLite, LangSmith, Guardrails AI

To-Do List MCP Server

Custom Model Context Protocol (MCP) server exposing task-management tools to LLM agents

- Built a fully functional MCP server in Python enabling AI agents to create, read, update, and delete tasks through structured tool calls.
- Implemented the server using FastMCP, applying the emerging MCP standard for connecting LLM agents to external tools and data sources.

Stack: Python, FastMCP, SQLite, uv

TECHNICAL SKILLS

Programming: Python, SQL
Generative AI: LLMs, RAG, Corrective RAG, LangChain, LangGraph, Prompt Engineering, MCP, Agentic AI
LLM Evaluation & Observability: RAGAS, LangSmith, Guardrails AI
ML / NLP: Scikit-Learn, TensorFlow, Keras, TF-IDF, Feature Engineering
Backend & Deployment: FastAPI, Streamlit, Docker, REST APIs, AWS
Databases: MySQL, SQLite, FAISS
Tools: Git, GitHub, Jupyter Notebook, n8n

EDUCATION

Master of Computer Science — Guru Nanak Khalsa College *2025 – 2027*

Bachelor of Information Technology —Bhavna trust College of commerce and science *2022 – 2025*

CGPA: 9.0 / 10

CERTIFICATIONS

- AI Engineering Bootcamp — Udemy (LangChain, RAG, Agentic AI, LangGraph)